GrowthPulse

SMART PLANT MONITORING

LoRaWAN Setup — Web App

GITHUB + NETLIFY (BRYAN)

Deploy the webhook and wire it into the app

GrowthPulse LoRaWAN Setup — Web App Checklist (Bryan)

Your job: get the LoRaWAN code deployed and wire the webhook into the app, so a node your teammate brings up in The Things Stack shows up in the app like any Wi-Fi node. Your teammate handles the gateway, The Things Stack, and the node join (separate checklist); you handle GitHub + Netlify.

Two things you need from your teammate (this is how the two halves connect):

1. Their TTS **device ID** (from step 4 of their checklist).
2. The agreed **webhook secret token** (any long random string you both use).

1. Protect main, work on a branch

```
cd "Senior Design 2/growthpulse-app"
git tag -a v2.12.2 -m "Stable before LoRaWAN" && git push --tags
git checkout -b lorawan-bringup
```

Now `main` is tagged and safe; if anything goes sideways, `git checkout v2.12.2`.

2. Commit + push the LoRaWAN files

These are already created in the repo:

- `netlify/functions/lorawan-uplink.js` (the webhook that lands uplinks in the app pipeline)
- `firmware/GP_LoRaWAN/GP_LoRaWAN.ino` (the node firmware your teammate flashes)
- `docs/GrowthPulse LoRaWAN Bring-Up Guide.(md/pdf)` and both checklists

```
git add . && git commit -m "LoRaWAN bring-up: webhook, bench firmware, guide" && git push -u origin lorawan-b
```

The webhook function goes live when this is deployed. You can deploy the branch as a Netlify deploy preview, or merge to `main` when you're ready for it on the live site. (Netlify auto-builds on push.)

3. Set the Netlify environment variables

Netlify → your site → **Site configuration** → **Environment variables**. Add:

Variable	Value
`LORAWAN_WEBHOOK_TOKEN`	the **secret token** you agreed with your teammate (must match the TTS webhook header exactly)
`LORAWAN_DEVICE_MAP`	JSON mapping their TTS device ID to a Losant device id, e.g. `{"their-device-id":"6a1fb486df527c8bf8d3324b"}`

Already set from before (confirm they exist): `LOSANT_APP_ID`, `LOSANT_COMMAND_TOKEN`, `VITE_SUPABASE_URL`, `VITE_SUPABASE_ANON_KEY`.

> `6a1fb486df527c8bf8d3324b` is your existing node's Losant device id. Use a different Losant device id if you want the LoRaWAN node to be a separate plant in the app. (If you'd rather not keep a map, you can instead set `LORAWAN_DEFAULT_DEVICE_ID` to a single Losant device id and skip `LORAWAN_DEVICE_MAP`.)

Trigger a redeploy after adding env vars (Netlify → Deploys → Trigger deploy) so the function picks them up.

4. Verify it end to end

Once your teammate's node is joined and uplinking, and your function is deployed with the env vars:

1. Netlify → **Functions** → **lorawan-uplink** → **logs**: you should see `forwarded: true` each time an uplink arrives (~every 60s in test).
2. Open the app → the mapped device should show live readings, and the connection badge should read **LoRaWAN** (the app shows the real link from the node's data).
3. If logs show `401 Bad webhook token` → the token in Netlify and the TTS webhook header don't match. `404 No Losant mapping` → the device id in `LORAWAN_DEVICE_MAP` doesn't match their TTS device ID.

5. When it works

Merge the branch to `main`:

```
git checkout main && git merge lorawan-bringup && git push
```

If you ever need to back out: `git checkout v2.12.2` returns you to the pre-LoRaWAN state.

You're done when: the Netlify function logs show forwarded uplinks and the LoRaWAN node appears live in the app. The next build step (merging LoRaWAN into the main firmware as a boot-selectable mode) happens after the join is proven, see Bring-Up Guide §8.